

Los bits de una instrucción

De la hoja de datos del procesador 8086 puede obtenerse la siguiente tabla:

Mnemonic and Description	Instruction Code			
DATA TRANSFER				
MOV – Move:	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0	7 6 5 4 3 2 1 0
Register/Memory to/from Register	1 0 0 0 1 0 d w	mod reg r/m		
Immediate to Register/Memory	1 1 0 0 0 1 1 w	mod 0 0 0 r/m	data	data if w = 1
Immediate to Register	1 0 1 1 w reg	data	data if w = 1	
Memory to Accumulator	1 0 1 0 0 0 0 w	addr-low	addr-high	
Accumulator to Memory	1 0 1 0 0 0 1 w	addr-low	addr-high	
Register/Memory to Segment Register	1 0 0 0 1 1 1 0	mod 0 reg r/m		
Segment Register to Register/Memory	1 0 0 0 1 1 0 0	mod 0 reg r/m		

NOTES:

AL = 8-bit accumulator

AX = 16-bit accumulator

CX = Count register

DS = Data segment

ES = Extra segment

Above/below refers to unsigned value

Greater = more positive;

Less = less positive (more negative) signed values

if d = 1 then "to" reg; if d = 0 then "from" reg

if w = 1 then word instruction; if w = 0 then byte instruction

if mod = 11 then r/m is treated as a REG field

if mod = 00 then DISP = 0*, disp-low and disp-high are absent

if mod = 01 then DISP = disp-low sign-extended to 16 bits, disp-high is absent

if mod = 10 then DISP = disp-high; disp-low

if r/m = 000 then EA = (BX) + (SI) + DISP

if r/m = 001 then EA = (BX) + (DI) + DISP

if r/m = 010 then EA = (BP) + (SI) + DISP

if r/m = 011 then EA = (BP) + (DI) + DISP

if r/m = 100 then EA = (SI) + DISP

if r/m = 101 then EA = (DI) + DISP

if r/m = 110 then EA = (BP) + DISP*

if r/m = 111 then EA = (BX) + DISP

DISP follows 2nd byte of instruction (before data if required)

*except if mod = 00 and r/m = 110 then EA = disp-high; disp-low.

if s w = 01 then 16 bits of immediate data form the operand

if s w = 11 then an immediate data byte is sign extended to form the 16-bit operand

if v = 0 then "count" = 1; if v = 1 then "count" in (CL)

x = don't care

z is used for string primitives for comparison with ZF FLAG

SEGMENT OVERRIDE PREFIX

0 0 1 reg 1 1 0

REG is assigned according to the following table:

16-Bit (w = 1)	8-Bit (w = 0)	Segment
000 AX	000 AL	00 ES
001 CX	001 CL	01 CS
010 DX	010 DL	10 SS
011 BX	011 BL	11 DS
100 SP	100 AH	
101 BP	101 CH	
110 SI	110 DH	
111 DI	111 BH	

Instructions which reference the flag register file as a 16-bit object use the symbol FLAGS to represent the file:

FLAGS = X:X:X:X:(OF):(DF):(IF):(TF):(SF):(ZF):X:(AF):X:(PF):X:(CF)

Ensamblar la siguiente instrucción para obtener todos los bits de la misma en lenguaje de máquina:

mov AX, 00720h

Modo de direccionamiento: Inmediato

Almacenar en un registro una palabra (word)

1 0 1 1 w reg | data | data if w = 1

if w = 1 then word instruction; if w = 0 then byte instruction

1011 1

reg = AX → 000

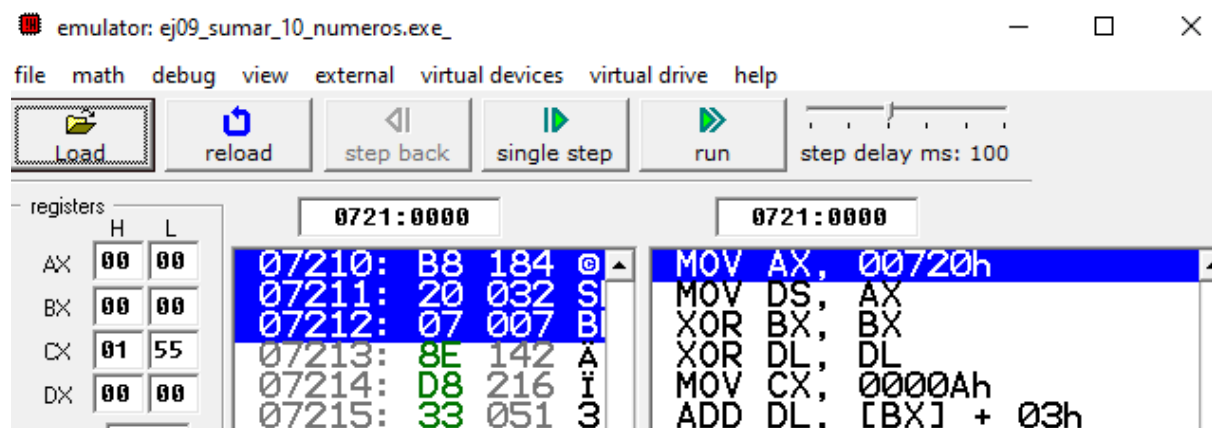
1011 1000

data if W = 1 little endian

0 7 2 0
0000 0111 0010 0000

La instrucción es:

1011 1000 0010 0000 0000 0111
B 8 2 0 0 7



Ensamblar manualmente la siguiente instrucción: `mov al, [27h]`